Spatial Localization in Visual Impairment

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Purpose. To investigate self-reported difficulties experienced by visually impaired subjects in real-world tasks requiring judgment of space and distance and to determine whether laboratory measures of spatial localization predict self-reported difficulty with spatial tasks better than traditional measures of visual function, such as visual acuity and contrast sensitivity.

Methods. Forty-two subjects with visual impairment participated. The Spatial Localization Questionnaire (SLQ) was developed to investigate self-reported spatial localization difficulties, and subjects answered the questionnaire as part of the study. Subjects also completed a variety of clinical vision tests (visual acuity, contrast sensitivity, stereo acuity, and reading speed) and laboratory vision tests (vernier acuity, bisection acuity, and visual direction).

Results. The SLQ was found to have good validity. Several significant correlations were found between the Rasch analysis ability scores for the questionnaire and the clinical and laboratory vision tests. Using stepwise regression analysis, we found that vernier acuity and contrast sensitivity accounted for 42% of the variance in the Rasch scores ($P < 0.001$).

Conclusions. The findings indicate that certain subjects with visual impairment have difficulty with real-world spatial tasks, as indicated by the SLQ. Of note, these difficulties were better predicted by vernier acuity (a resolution test) and contrast sensitivity, rather than vernier or bisection bias, which measure localization. (*Invest Ophthalmol Vis Sci.* 2006;47:78–85) DOI:10.1167/iovs.05-0137

Many people with visual impairment complain about difficulties with activities requiring judgment of space and depth. Examples include negotiating steps and pouring liquids. It is well known that subjective complaints of difficulty are not necessarily well correlated to objective measurements.[1,2] For example, difficulties with activities of daily living are not well correlated with clinical measures of visual acuity.[3–5] In particular, difficulties with spatial tasks do not appear to be related to the level of visual acuity and visual field loss,[6,7] but performance may depend on an individual's ability to localize objects in the surrounding visual world. This ability is commonly referred to as spatial localization and has traditionally been measured using hyperacuity tasks such as vernier acuity, bisection acuity, spatial interval discrimination, and stereo acuity.[8,9] Spatial localization has been investigated in retinitis pigmentosa,[6,7] glaucoma,[10,11] and amblyopia.[12–15] It has also been used as a method of determining potential acuity in cataract[16] and to screen for metamorphopsia in early age-related macular disease (ARM).[17] Turano and Schuchard[6] measured spatial localization in visually impaired individuals with central and peripheral visual field loss and found that poor spatial localization is not a necessary consequence of visual field loss, since some subjects with visual field loss made spatial relationship judgments with an accuracy comparable to that of subjects with normal vision.

Several questionnaires and instruments have been developed specifically for use with a visually impaired population to identify difficulties with functional vision. Some notable examples are the National Eye Institute Visual Function Questionnaire (NEI-VFQ),[18] The Daily Living Tasks Dependent on Vision (DLTV),[19] Low Vision Quality of Life Questionnaire (LVQOL),[20] Melbourne Low Vision ADL Index (MLVAI),[21] and Veterans Affairs Low Vision Visual Functioning Questionnaire (VA LV VFQ-48).[22] Although many of these questionnaires have items related to difficulties with spatial tasks, none of the existing questionnaires is exclusively designed to identify such difficulties. There was therefore a need to develop a questionnaire that would specifically ask questions that relate to the difficulties experienced with tasks requiring judgments of space and depth. This questionnaire could be used to identify visually impaired individuals who experience functional difficulties in tasks that require distance or space judgments. These difficulties could then be correlated to clinical and laboratory tests to identify which tests best predict spatial localization ability. The results of those tests that best predict such difficulty could then be used clinically to give appropriate advice to patient and plan rehabilitation services.

Vernier and bisection acuity quantify an individual's ability to determine relative location, and these measures were therefore chosen for this study, to objectively assess relative location abilities. For certain tasks such as reaching and grasping, however, this information may be insufficient, and the subject will also require information about absolute direction. No previous studies have addressed the judgment of "straight ahead" in the visually impaired and methods used for normally sighted individuals[23,24] may not be suitable. It is known that perceived visual direction is shifted in the direction opposite to that of eccentric gaze,[25] and this phenomenon is particularly relevant in patients with a central field defect, who consistently use an eccentric PRL to replace the fovea. An experiment was therefore designed to measure visual direction. Accurate perception of object size also appears to be relevant to performance of these activities. It is well known that for normally sighted subjects, the perceived size of an object does not change when altering the viewing distance between the object and the observer, despite the fact that there is a change in retinal image size (size constancy).[26,27] Size constancy, however, has been found to be normal in a range of visually impaired individuals[28,29] and so was not included in this study.

The purpose of this study was to investigate possible correlation between visual function and self-reported real-world spatial localization difficulties in visually impaired individuals. We hypothesize that laboratory measures of spatial localization will predict self-reported difficulty with spatial tasks much better than traditional measures of visual function, such as visual acuity and contrast sensitivity.

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METHODS

Subjects

Visually impaired subjects who participated in the study were recruited either from the Low Vision Clinic at the Manchester Royal Eye Hospital or the Low Vision Clinic at University of Manchester Institute of Science and Technology (UMIST). All the subjects from the Manchester Royal Eye Hospital had been referred to the Low Vision Clinic by their ophthalmologists. Subjects who were recruited from UMIST were referred by optometrists, ophthalmologists, or rehabilitation workers or were self-referred. Visually impaired individuals were recruited for the study only if they had visual acuity between 6/6 and 6/60 (0.0–1.0 logMAR [logarithm of the minimum angle of resolution]) in the better eye and no better than 6/9.5 (0.2 logMAR) in the poorer eye and had attended an appointment at a low-vision clinic. Although some subjects recruited in the low-vision category had better visual acuity than would normally be categorized as visually impaired, all these subjects had attended a low-vision clinic complaining of visual problems despite their good vision, suggesting that they experienced or perceived difficulty that was not in proportion to their measured visual acuity loss. Subjects who lived in a hospital or nursing home and/or did not speak English were excluded. Subjects were also excluded if they failed to achieve 17 correct responses (of a possible 22) on the Mini Mental State Examination for the Blind (MMBlind).[30]

Forty-two individuals participated in the study, although only 40 completed all the experiments. Two subjects missed the second visit because of ill health.

Testing Procedures

Most of the subjects ($n = 39$) completed the experimental test procedure over two sessions (median duration between sessions, 2 weeks; range, 1–8 weeks) and the remaining subject completed the tests in one session. Visual acuity, at both distance and near, and contrast sensitivity measurements were tested on the first visit. The spatial localization questionnaire (SLQ) was also administered to all subjects ($n = 42$) on the first visit. Visual acuity at distance was repeated at the second visit, to check that the vision remained stable between visits. None of the subjects had a clinically significant change in visual acuity (difference of greater than 0.1 logMAR) between visits.[31] The remainder of the experiments were randomly divided between the two visits.

An objective and subjective refraction was performed on all subjects, and all subsequent testing was performed using the best correction for the viewing distance.

Ethics committee approval for the study was obtained from the Central Manchester Local Ethics Research Committee (Reference Number: CEN/00/111), and the study adhered to the tenets of the Declaration of Helsinki. Before participating in the study, all individuals gave informed consent.

Spatial Localization Questionnaire

A questionnaire was developed to investigate practical difficulties relating to judgments of space and distance at close range in visually impaired subjects. Possible questions were either short listed from previous questionnaires[18–21,32–36] or were based on the investigators' experience. A list of 233 potential questions was constructed, of which 39 were retained for investigation. These questions were pilot-tested on a sample group of 20 elderly, visually impaired individuals attending a low-vision clinic. Based on subject response and feedback (e.g., questions that required additional explanation or were applicable to very few subjects) the questionnaire was further reduced. The final version of the SLQ consisted of 30 questions rated on a five-point scale (5, no difficulty; 4, slight difficulty; 3, moderate difficulty; 2, severe difficulty/regularly uses sensory substitution; 1, cannot do task because of vision/gets someone else to do it; NA: has never done task/cannot do task because of reasons unrelated to vision). All the SLQ items started with the phrase "How much difficulty do you have. . . ." The SLQ was validated using Rasch analysis,[37] which is based on a probabilistic model and has recently been used to validate several vision-related quality-of-life questionnaires.[21–22,38–41] The Rasch analysis was performed with a joint maximum likelihood estimation (student version of Winsteps, ver 3.33; Mesa Press, Chicago, IL). This analysis can be used to transform raw data into abstract equal-interval scales and to measure construct validity.[42] Rasch analysis may require a reduction in the instruments' items and response categories for the data to fit the model. It should be noted that the questionnaire was administered to 42 subjects, which may be too small a sample size for assessing the psychometric properties of a new questionnaire. The questionnaire took no more than 15 minutes to administer.

Clinical Vision Tests

Visual acuity at distance was measured with the Early Treatment Diabetic Retinopathy Study (ETDRS) chart.[43] Reading speed was assessed with the MNREAD acuity charts and established protocols.[44] Contrast sensitivity was measured using the Pelli-Robson contrast sensitivity chart.[45] All these measures were performed binocularly and monocularly. Stereopsis was tested using the Frisby Stereo Test.[46,47] Visual fields were not measured as part of the study.

Laboratory Vision Tests

All targets for the vernier and bisection acuity experiments were generated in a program written in C++ with Delphi and were projected by a data projector (model CP-X320W; Hitachi, Tokyo, Japan) onto a white screen that was 4.5 m away from the subject. System resolution was 1024×768 pixels, and 1 pixel subtended 45 seconds of arc at the subject's eye. The projector was fixed at a distance of 2 m from the screen and was not in the subject's line of sight. The luminance of the target was 300 cd/m^2, and the background was 105 cd/m^2. Testing was binocular with the room lights switched on, giving illuminance of 800 lux. The method of constant stimuli is usually used because of decreased variability, but because it requires several hundred trials to determine threshold, the method of adjustment was used to avoid subject fatigue.

Vernier Acuity

The vernier target that was used consisted of two bright white vertical lines, $3° \times 1°$ in size, placed one above the other and separated by a gap of 1°, on a dark background. The top part of the target was fixed and the bottom part was positioned either to the left or right. The subject was required to use the keyboard to move the bottom target so that it was aligned with the top target. The subject received no feedback on performance. The examiner randomly displaced the bottom target either left or right before each threshold determination. The experiment was repeated 17 times, and the first two trials were discarded to account for any learning effects. Each set of 17 trials took no more than 10 minutes to perform. If the bottom line was displaced to the right, the displacement was considered positive, and if it was displaced to the left, the displacement was considered negative. The mean of the vernier displacements was taken to be the vernier bias, which indicates accuracy or mean error of alignment, and the SD was taken to be the vernier acuity.

Bisection Acuity

Bisection acuity was measured to determine whether subjects had any difficulty in dividing a spatial interval into two equal parts. The bisection target consisted of a horizontal bright white line with a vertical bright white cursor on a black background. Two different lengths of horizontal line were tested (1° and 3°) with a standard width of 20 minarc for both. No references were found in literature that suggested ideal line lengths particularly for visually impaired subjects. When assessing the optimum performance of normally sighted subjects, this experiment is normally conducted as a three-line bisection task requiring the observer to judge when a central line is an equal distance from two outer lines with a separation of only a few minutes of arc (e.g.,

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Wang et al.[48]). The larger target used in this experiment was chosen to be appropriate to poor acuity and to represent the scale of real-life practical tasks. The size of the vertical cursor used was always the same (1° × 20 minarc), regardless of the horizontal line length tested. The subject viewed a horizontal line on the screen with a vertical cursor displaced randomly, either to the right or the left of the center of the horizontal line. The subject had to use the computer keyboard to move the cursor so that it divided the line into two equal halves. A total of 17 trials were performed for both the line lengths, and the first two trials for each line length were discarded. The order of performing the experiments was randomized so that some subjects performed the experiment with the 3° line first, whereas others performed the experiment with the 1° line first. Each experiment took no more than 10 minutes to perform. Similar to vernier acuity, the mean of the bisection measurements was taken to be the bisection bias, and the SD was taken to be the bisection acuity.

Visual Direction

The experimental setup consisted of an optical bench in the frontoparallel plane with a movable target mounted on it. The target was a red LED (2 mm × 1 cm) enclosed in a black surround that could be moved using two strings. The distance between the subject and the target was 60 cm, and the target was at eye level. The subject's head was stabilized with a chin rest with a special attachment for the subject to place his or her nose. The apparatus was constructed so that the subject would not move his or her head. It was assumed that the veridical "straight ahead" was an imaginary line drawn from the subject's nose to the target. A millimeter scale was attached to the optical bench and the assumed position of "straight ahead" was marked as zero. Subjects viewed the target through a window constructed using matt-black card so that only the target was visible. The background was also black. At the start of each experiment the examiner displaced the target either left or right of the central position. The subjects were instructed to move the target using the two strings (which were at the side of the table) until they felt that the target was straight in front of them. Subjects were instructed to make judgments with both their eyes open and under no circumstances to close one eye. Strings were used to adjust the target so that the patient could not use any information from hand position to judge visual direction. If the subject was unable to see the red LED (and five subjects could not), they were instructed to look at the black surround (4 cm × 2.5 cm, angular measure of 3.81° × 2.38° at 60 cm) enclosing the LED. Five readings were taken per subject, which did not take more than 5 minutes to perform. The mean of the visual direction measurements was taken to be the visual direction bias, and the SD was taken to be the visual direction ability.

Results

All statistical tests were performed on computer (SPSS, ver. 10.0; SPSS Inc., Chicago, IL).

Subject Demographics

The mean age of the visually impaired subjects was 73 years (range, 36–89 ± 12.5 years [SD]). Of the subjects, 76.2% (n = 32) had bilateral ARM, 9.6% (n = 4) had bilateral idiopathic optic atrophy, 4.7% (n = 2) had bilateral ARM with glaucoma, 4.7% (n = 2) had bilateral diabetic retinopathy, 2.4% (n = 1) had bilateral macular holes, and 2.4% (n = 1) had bilateral resolved cystoid macular edema. The median duration of the ocular disease was 3 years (range, 1–60) in the right eye and 3.5 years (range, 0.5–80) in the left eye. The best corrected binocular visual acuity of the subjects was 0.47 ± 0.30 logMAR.

Spatial Localization Questionnaire

The Rasch model estimates item difficulty and person ability as log odds ratios (logits). In Table 1, item difficulty refers to the logit item measure, and "error" refers to the standard error estimate for each item. Positive item measures imply that items require better localization ability than the average items and negative item measures imply that items require less localization ability than average items. The fit of the items is assessed either using the infit mean square (information-weighted fit) or the outfit mean square (outlier-sensitive fit) statistic which compares the differences between the predicted and observed responses. Alternatively, the outfit and the infit ZSTD can be used, which are the normalized infit and outfit mean squares expressed in model SD units. The ZSTDs were used to assess fit in the present study. For an item to fit the model well, it is usually assumed that the ZSTD should be between +2 or −2 SD units.[40] Rasch analysis was initially performed on the 30-item SLQ. Items 8 and 23 lay outside the acceptable ZSTD range, however, because the ZSTD statistics were very close to the acceptable range it was decided to retain these items for subsequent analysis. Items 16 (fastening buttons) and 25 (playing bingo) were considered redundant based on the fact that only seven subjects had difficulty with item 16 and only five found item 25 applicable, and it was therefore decided to delete these items. Analysis with the reduced 28-item questionnaire resulted in a subject reliability index of 0.95 and item reliability index of 0.94. Rasch analysis on the reduced item SLQ is summarized in Table 1. Again, the ZSTDs for items 8 and 23 were very close to the acceptable ±2 ZSTD tolerance, suggesting that the misfit of these items can be attributed to noise rather than anomalous responses by some subjects. It was therefore decided to retain these items in the SLQ. The person–item map for the reduced item SLQ is illustrated in Figure 1. The vertical line is a logit scale and represents localization ability. Persons appear on the left-hand side of the scale in ascending order of ability, with the more able subjects appearing at the top of the scale and the less able appearing at the bottom. Items appear on the right-hand side of the scale in ascending order of difficulty, with the more difficult items being at the top of the scale and the least difficult items appearing at the bottom. Analysis of the person–item map indicates that there are too few items at the top of the scale, which implies that the instrument would be unable to distinguish between people with high ability. Also, many of the items are clustered together, indicating a redundancy based on the fact that they are measuring the same ability.

Rasch analysis can be used to determine whether sufficient numbers of category responses (rating scales) have been used or whether categories need to be added or merged with other categories. In the present study, a small sample size was used to validate the questionnaire, and in such circumstances it is extremely difficult to make decisions about the use of response categories.[49] It was therefore decided to retain all five categories in the analysis, and category response curves were not calculated. Rasch ability scores (in logits) for the SLQ (26-item, five-point rating scale) were used for subsequent data analysis, since these scores were interval scores. These scores ranged from −1.58 to 3.41, with a mean value of 0.58 ± 1.24 (SD).

Relationship between Visual Function Tests and SLQ

The means and standard deviations for the clinical and laboratory vision tests are listed in Table 2. To determine which variables may be predictive of subjective real-world spatial tasks, stepwise regression analysis was performed using 11 variables (distance acuity, reading speed, contrast sensitivity, vernier acuity and bias, bisection acuity and bias (1° and 3° targets), visual direction, and visual direction ability) and only binocular scores were included. Because stereopsis was measurable in only five subjects, it was not included in the regression analysis. The SLQ person–ability scores were the depen-

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TABLE 1. Rasch Analysis on the Reduced Item SLQ

Item Number	Item Details	Count	Item Difficulty (Logits)	Error	Infit MNSQ	Infit ZSTD	Outfit SQ	Outfit ZSTD
26	Threading a needle	39	2.81	0.25	1.26	0.8	1.32	0.7
27	Sewing a button	28	1.61	0.24	1.44	1.3	1.49	1.2
4	Writing in straight lines	42	0.62	0.17	0.84	−0.8	1.06	0.2
28	Using a screwdriver	33	0.57	0.19	1.07	0.3	0.95	−0.2
20	Crossing the road	42	0.5	0.17	0.99	−0.1	1.25	0.9
29	Using a hammer	32	0.48	0.19	0.6	−1.8	0.54	−1.8
30	Writing in crossword boxes	30	0.47	0.19	1.45	1.6	1.33	1.0
6	Measuring cooking ingredients	27	0.41	0.21	1.56	1.8	1.52	1.3
10	Knob setting on appliances	39	0.31	0.18	0.9	−0.4	0.76	−0.9
2	Squares on forms	42	0.28	0.17	1.12	0.6	1.15	0.5
24	Changes in ground level	42	0.05	0.17	1.04	0.2	1.12	0.4
1	Telephone buttons	42	0.02	0.17	0.9	−0.5	0.82	−0.6
14	Inserting plugs	42	−0.01	0.17	0.96	−0.2	0.78	−0.8
15	Inserting key into keyhole	42	−0.04	0.17	1.07	0.3	0.92	−0.3
17	Judging when zipper is in place	42	−0.04	0.17	0.99	0	0.89	−0.4
9	Microwave buttons	32	−0.13	0.2	1.15	0.5	1.05	0.1
5	Judging when a cup is full	42	−0.16	0.17	1.07	0.3	1.24	0.7
19	Putting coins into a slot	37	−0.2	0.19	0.75	−1.1	0.61	−1.2
13	Changing batteries	39	−0.26	0.18	1.01	0	0.84	−0.4
21	Judging the next stair	42	−0.28	0.17	0.91	−0.4	0.92	−0.2
3	Signing name on a check	42	−0.4	0.18	0.87	−0.6	0.83	−0.5
18	Shaving/applying makeup	38	−0.49	0.18	1.31	1.1	1.55	1.1
22	Avoiding obstacles	42	−0.62	0.18	0.81	−0.8	0.74	−0.7
7	Judging food on a plate	42	−0.75	0.18	0.88	−0.5	0.77	−0.6
11	Finding utensils in a closet box	41	−0.97	0.19	1.29	1.1	1.35	0.6
23	Walking in a straight line	35	−1.18	0.22	1.8	2.2	2.31	1.5
12	Tops on bottles/jars	42	−1.19	0.2	0.71	−1.2	0.53	−1.0
8	Quantity of food to put on plate	40	−1.42	0.21	0.49	−2.2	0.42	−1.2
	Mean	39	0	0.19	1.04	0	1.04	0
	SD	5	0.84	0.02	0.29	1	0.39	0.9

Count refers to the total number of subjects who answered an item. Item difficulty refers to the estimate of visual ability required by the item. Error refers to the standard error of the estimate of item measure. Infit MNSQ refers to the infit mean square. Outfit MNSQ refers to the outfit mean square.

dent measure. Stepwise regression analysis revealed that two variables— contrast sensitivity and vernier acuity—accounted for 42% of the variability observed in the questionnaire score ($r = 0.64$, $r^2 = 0.42$; $P < 0.0001$). Pearson's correlation coefficients were also calculated for each of these variables with the SLQ and are listed in Table 2. The regression coefficients (B) with their standard errors for the regression equation are summarized in Table 3.

DISCUSSION

It is generally accepted that some visually impaired subjects have more difficulty performing activities of daily living than is suggested by their visual acuity (see, for example, Rubin et al.[50]). In the particular case of real-world tasks that require the judgment of depth and space, no study has been attempted to explain why certain visually impaired subjects have difficulties with these tasks and what visual factors might be predictive of these difficulties. In the present study, we developed a questionnaire that enabled us to discriminate between different levels of self-reported difficulty experienced in such tasks by visually impaired individuals.

The results of the Rasch analysis indicate that the SLQ is a valid questionnaire, as the infit and outfit statistics seem favorable, with very little noise and dependency in the data, with the exception of items 8 (quantity of food to put on plate) and 23 (walking in a straight line). These items may be misfitting, because subjects may have misunderstood the question or may have confounded the question with other factors. For example, walking in a straight line may have been confounded with physical ability. The easiest item on the questionnaire was fastening a button. This is not a surprising finding, as subjects may be able to do this task by touch, irrespective of the level of acuity and precision. In contrast, threading a needle requires good acuity and precision; and, unsurprisingly, this was the most difficult item on the questionnaire. As discussed in the results section, the SLQ may not distinguish between subjects at the higher end of the ability scale. It can be argued that questions are not required that would distinguish between people with high ability, because such persons do not have any functional difficulty and therefore do not need any help. Questions are required that distinguish people with low and medium levels of ability and the items in the questionnaire seem to cover this range. Although the SLQ was found to be a valid measure, it was validated on a small sample in subjects mainly affected by ARM (32 subjects who took part in this study had ARM), who were specifically chosen for the study. In the future, the SLQ should be validated on a larger, more random visually impaired population and test–retest values need to be established.

The main hypothesis in the present study was that laboratory measures of spatial localization would predict self-reported difficulty with spatial tasks much better than traditional clinical measures of visual function, such as visual acuity or contrast sensitivity. The laboratory measures investigated were vernier acuity and bias and bisection acuity and bias. We thought that measures of vernier and bisection bias would predict self-reported difficulties with spatial tasks better than vernier and bisection acuity; because, theoretically, bias or constant error should correlate with spatial localization, whereas precision or variable error (vernier and bisection acuity) should correlate with resolution. In the present study,

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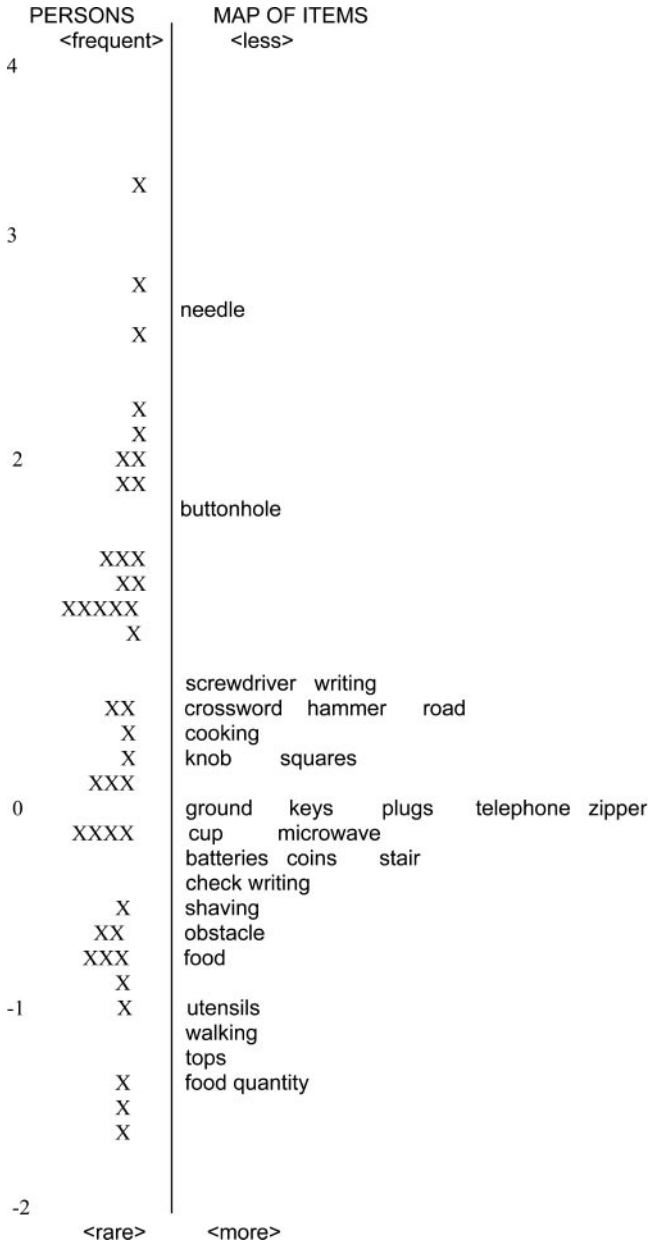


FIGURE 1. Person ability/item difficulty map for the 26-item SLQ. *Left of vertical line*: persons (represented by X). *Right*: items. Harder items and more able people are at the *top* of the scale, and easier items and less able people are at the *bottom*. The scale is in log units (logits).

the only bias measure that correlated significantly with self-reported spatial localization difficulties was vernier bias. The other measures of bias did not correlate significantly. Measures of precision, however, seemed to be better correlated with the spatial localization score, with both vernier acuity and bisection acuity for the larger target size correlating significantly. When a stepwise regression analysis was performed, it was found that vernier acuity and contrast sensitivity were the two variables that best predicted the spatial localization score. It appears that visually impaired subjects, particularly those subjects with ARM, have greater difficulty with precision judgments and may simply be more variable at performing tasks, and this variability may explain why they have difficulty with tasks that require judgments of space and depth. Turano and Schuchard[6] and Turano[7] found localization defects in individuals who had from retinitis pigmentosa, and these defects seemed to be independent of visual acuity and visual field loss. The finding of the present study that changes in contrast sensitivity correlate significantly with the variability in the questionnaire score is similar to results in several previous studies. It has been shown that contrast sensitivity correlates significantly with various tasks of daily living, such as reading performance,[51–53] mobility,[54] and perception of faces[55] and with difficulties in tasks requiring distance judgments.[3]

Although there was a small but significant correlation between the questionnaire score and visual acuity, other measures of visual function were better predictors of self-reported difficulty with real-world spatial tasks. Other investigators have found a similar lack of correlation between acuity and practical tasks, such as mobility,[52] in individuals who are visually impaired. The low association between the questionnaire score and visual acuity in this study, however, may be due to the low-vision population that was selected. For this study, individuals were selected who had fairly good visual acuity, but they had complaints that were not thought to be explained by visual acuity alone. The sample that was selected for this study was therefore not a typical low-vision sample and, had a more typical sample been selected, a better association may have been found. Rubin et al.[3] studied a random sample of 220 community-dwelling adults in the United States and found that visual acuity was associated with difficulty in tasks requiring good resolution and changing light conditions. Other investigators[40,56–60] have also found good correlation between visual acuity scores and ability to perform activities of daily living. For example, Gothwal et al.[59] administered the L. V. Prasad-Functional Vision Questionnaire (LVP-FVQ) to 78 visually impaired Indian school children and reported a Pearson correlation coefficient of -0.57 between the person ability scores and logMAR visual acuity. The LVP-FVQ was also administered to 128 visually impaired adults, and a similar correlation was found between the person ability scores and LogMAR visual acuity.[60] Massof and Fletcher[40] also showed a good correlation ($r = -0.53$) between the NEI-VFQ person–ability scores and visual acuity. These investigators used a larger and more random sample of visually impaired subjects and that may explain in part the difference in results.

Only approximately half the variability observed in the questionnaire score was attributable to the visual factors chosen for this study. This may be because this study selected only a range of visual factors that could be measured in a clinic or a laboratory. The activities of daily living on which the questionnaire is based are complex tasks and are also dependent on many nonvisual factors, such as levels of motivation, personality, and social support. Many investigators have speculated that these factors play an important role in the performance of day-to-day activities (see for example Szlyk et al.,[61] Rosenbloom,[62] Davis et al.,[63] and Jang et al.[64]). Physical and mental health,[65] age, and manual dexterity may also have accounted for some of the variability in the questionnaire scores. Some subjects had arthritis, although none thought that it was severe enough to limit them physically. It is also possible that visual field loss correlates significantly with self-reported localization difficulties; however, visual fields were not measured in this study, because we thought that the available means of testing visual fields, such as automated perimetry, Amsler grid or the tangent screen, would not yield reliable results, particularly since it has been shown that patients with a central scotoma often fixate with a preferred retinal locus instead of the fovea, despite verbal instructions being given.[66]

Possible mechanisms for localization defects in individuals who are visually impaired have not been investigated. A vernier acuity loss in persons with amblyopia is well reported and suggested to be due to undersampling[67] or neural disarray.[68] According to the undersampling theory, because of neuronal

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TABLE 2. Results of the Clinical and Laboratory Vision Tests

Clinical Test	*n*	Mean	SD	Correlation with SLQ	*P*
Visual acuity (logMAR)	42	0.47	0.30	−0.38	0.01
Contrast sensitivity (log units)	42	1.20	0.30	0.53	<0.01
Reading speed (words per minute)	40	143.40	49.32	0.53	<0.01
Stereopsis (sec of arc)	5	268.75	225.03	−0.26	0.08
Vernier acuity (min of arc)	42	188.88	109.33	−0.59	<0.01
Vernier bias (min of arc)	42	45.74	304.36	−0.35	0.02
Bisection acuity (1° target, min of arc)	40	116.24	60.96	−0.23	0.15
Bisection bias (1° target, min of arc)	40	29.34	82.57	−0.05	0.76
Bisection acuity (3° target, min of arc)	42	245.05	140.74	−0.27	0.08
Bisection bias (3° target, min of arc)	42	68.82	298.48	0.03	0.83
Visual direction (cm)	42	0.067	1.33	−0.11	0.48
Visual direction ability (cm)	42	0.390	0.19	−0.24	0.13

All tests were performed binocularly.

loss, subjects with amblyopia do not have enough neurons to be able to determine object position with certainty. Although the origin of loss is presumed to be cortical, it is conceivable that the cellular loss at the level of the retina could help explain the presence of localization defects in visually impaired subjects. Most of the visually impaired subjects who took part in our research had ARM, and it is associated with a loss of visual cells due to the degeneration of RPE cells.[69] This loss could quite easily result in a sparse number of cells, which might explain poor positional ability in visually impaired individuals, especially individuals who have macular degeneration. According to the neural-disarray theory, spatial position "tags" or neural signals may be altered or scrambled so that the cellular response is mislabeled, coming from the wrong position in space. As with the undersampling theory, the origin is presumed to be cortical. However, it is conceivable that in a disease such as ARM, photoreceptors could be moved out of position by hemorrhage and or drusen resulting in an alteration or scrambling of signals. An alternative third explanation could be that poor localization may simply be due to visual field defects, particularly if these defects are in the central visual field. Eccentric viewing caused by a central scotoma would also lead to the use of peripheral retina, which has poorer localization abilities.[70]

It seems plausible that the duration of the visual impairment would play an important role in determining level of difficulty; however, it was not possible to determine accurately the onset of vision loss in most of the subjects who participated in the study. The onset of vision loss reported in this study was based on the subject's memory, which may not have been accurate. Many subjects reported that they could not remember which eye had been affected first and were reporting approximate values for onset. One would expect that the longer the duration, the more likely it would be that the individual would have adapted to the visual loss. Several of the visually impaired subjects in the study had one eye that was more severely affected, and some of these subjects reported that they had great difficulty in judging distances and depth at the onset of their vision loss; however, they were able to adapt to their loss within a few months. Most of the subjects reported stable vision for several months before this study. It may be that if vision is constantly changing, individuals experience greater difficulty, since they have to readapt constantly to their new visual status.

TABLE 3. Multivariate Regression Model

Variable	B	Std. Error
Constant	−0.28	0.92
Vernier acuity	−4.8E-03	0.01
Contrast sensitivity	1.43	0.59

Prediction of self-reported localization difficulty from vernier acuity and contrast sensitivity.

An interesting aspect of this study was that only 5 of the 42 visually impaired subjects had measurable stereo acuity on the Frisby test. Rubin et al.[71] tested individuals using the Randot Circles test (Stereo Optical, Chicago, IL) and found that elderly individuals who have four or more lines of difference (0.4 logMAR or greater) in visual acuity or 0.60 log units or more of difference in contrast sensitivity between the two eyes, lacked stereo acuity. All the five individuals who had measurable stereo acuity in the present study had acuity differences of less than 0.4 and contrast differences of less than 0.6 log units between the two eyes. Nine other subjects also had similar acuity and contrast differences, yet did not demonstrate measurable stereo acuity. Most of these nine subjects had only slight to moderate difficulty in performing localization tasks, and it was surprising to note that they lacked stereo acuity. It is hard to establish what could be the cause of this, even after accounting for the fact that all the subjects who took part in the present study were elderly and that stereo acuity declines with age.[72,73] One possibility is that subjects were not able to see the pattern on the Frisby test plates, but that seems unlikely, because the Frisby test plates are of high contrast and all the subjects who participated in the study reported that the patterns on the Frisby plates were visible. Likely causes for this decline should be investigated further.

The SLQ can be used to measure self reported difficulty with real world localization tasks. The questionnaire takes less than 15 minutes to administer and does not involve complicated equipment. Information can also be gained by measuring contrast sensitivity, which would probably be the preferred method of choice for clinicians. Vernier acuity testing requires more sophisticated instrumentation that limits its use to a laboratory; however, it may be possible to design a simple vernier test that can be used in a clinical setting. Further research should be undertaken to explore this possibility.

Various studies have shown that visual tasks such as vernier acuity improve with practice.[74,75] However, most of these studies have found that training an individual on one task does not necessarily lead to improvement on another task, and many practice sessions are required before thresholds improve (see, for example, Fendick and Westheimer[76]). Therefore training for visually impaired subjects would probably be best directed to the task itself rather than the underlying visual function. Another practical recommendation for individuals with spatial

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localization difficulties would be to improve the contrast of the task.

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